

Week 7 Tutorial | Games, Adversarial Search

Name _____

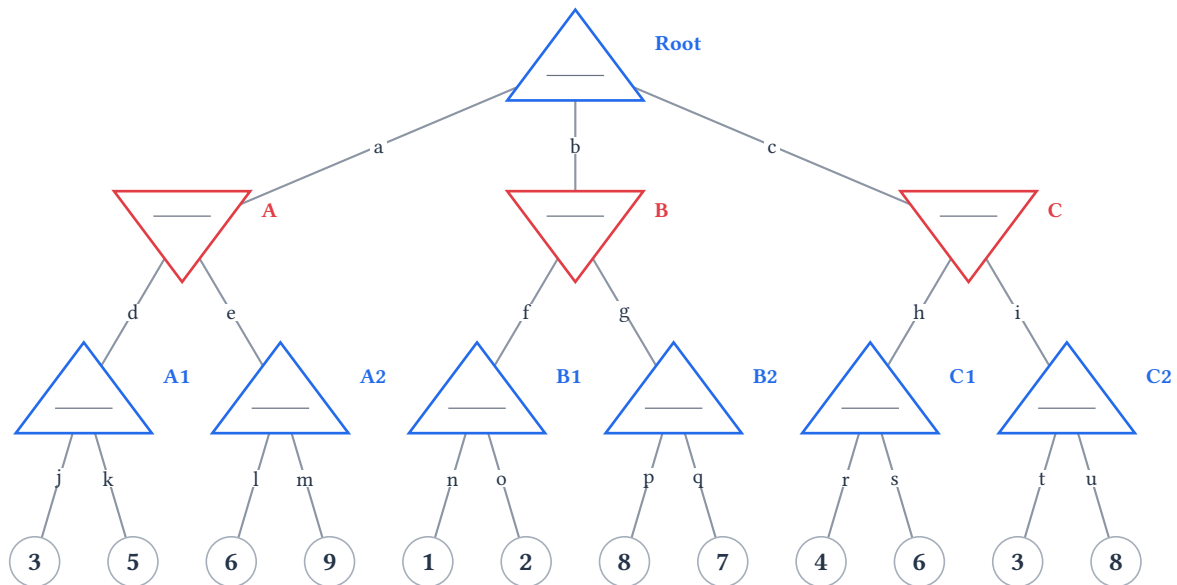
Week 7

Team name: **Triple L**

(a) Minimax

Fill in **all internal node values** on the tree, working from the leaves upward. State the root value and the best root action (a, b or c).

△ MAX ▽ MIN



Root value: _____

Best root action: _____

(b) α - β pruning

Start a **fresh search**, visiting children **from left to right**, with $\alpha = -\infty$ and $\beta = +\infty$ at the root. Do not use values calculated in (a) to skip ahead.

Cross out every unvisited part of the tree. At each cutoff, label the node and the current α and β . Count only leaves actually evaluated.

Cutoff 1: _____ $\alpha =$ _____ $\beta =$ _____

Cutoff 2: _____ $\alpha =$ _____ $\beta =$ _____

Leaves evaluated: _____ / 12

(c) Chance nodes

Replace A, B and C with **CHANCE** nodes. Each chooses its two **MAX** children with probability $\frac{1}{2}$ each. Keep the root and the lower **MAX** nodes unchanged. Use the **full tree**: recalculate the three values and the root's best action(s).

$V(A) =$ _____ $V(B) =$ _____ $V(C) =$ _____

Best root action(s): _____ Does the choice change? _____

Answers

Team name: Triple L

(a) Minimax

Node A · action a	Node B · action b	Node C · action c
$A1 = \max(3, 5) = 5$	$B1 = \max(1, 2) = 2$	$C1 = \max(4, 6) = 6$
$A2 = \max(6, 9) = 9$	$B2 = \max(8, 7) = 8$	$C2 = \max(3, 8) = 8$
$A = \min(5, 9) = 5$	$B = \min(2, 8) = 2$	$C = \min(6, 8) = 6$

The root value is $\max(5, 2, 6) = 6$. MAX chooses action c (leading to C).

(b) α - β pruning

1. Cutoff at A2 (MAX). A1 returns 5, so A passes $\beta = 5$ to A2. After A2 reads its first leaf, 6, its bound is $\alpha = 6$. Since $\alpha \geq \beta$ ($6 \geq 5$), skip the leaf 9. A already has an alternative worth 5 and will not choose A2, whose value is at least 6.

2. Cutoff at B (MIN). A returns 5, so the root passes $\alpha = 5$ to B. B1 returns 2, making $\beta = 2$ at B. Since $\beta \leq \alpha$ ($2 \leq 5$), skip all of B2, including leaves 8 and 7. B can offer at most 2, so it cannot improve the root's current choice.

Finish C. C1 returns 6; C2 reads 3 and then 8. Both leaves are visited, so no additional leaves are pruned. C returns 6, so c becomes the best root action.

Visited, in order: 3, 5, 6, 1, 2, 4, 6, 3, 8.

Total: 9 / 12 leaves evaluated; 3 leaves pruned. Root value: 6, action: c.

The cutoff at A2 establishes a **lower bound** of 6, not its exact minimax value of 9. B2 is never evaluated. α - β can select the correct root action without finding every internal node's exact value.

Changing visit order can change the amount of pruning, but not the root minimax value. It may change which equally good action is selected when there is a tie.

(c) Chance nodes

The lower MAX values stay the same. At each CHANCE node, take the probability-weighted average:

$$V(A) = \frac{1}{2}(5) + \frac{1}{2}(9) = 7, \quad V(B) = \frac{1}{2}(2) + \frac{1}{2}(8) = 5, \quad V(C) = \frac{1}{2}(6) + \frac{1}{2}(8) = 7.$$

The root value is 7: actions a and c tie (nodes A and C). Action c is still optimal, but it is no longer the unique best action.

MIN models an opponent choosing the worst branch for MAX. CHANCE models an outcome drawn according to given probabilities, so we take an expectation. This modified tree is **expectimax**; a tree containing MAX, MIN and CHANCE nodes uses **expectiminimax**.

The earlier cutoffs relied on MIN's choice. Do not reuse them for this chance calculation: the skipped branches contribute to the expectation.